

Midterm Exam Solutions

(5:35 PM – 6:50 PM : 75 Minutes)

- This exam will account for 25% of your overall grade.
- There are five (5) questions, worth 75 points in total. Please answer all of them in the spaces provided.
- There are 18 pages including 9 blank pages. Please use the blank pages if you need additional space for your answers.
- This is a *closed book, closed notes* exam. *No cheat sheets* are allowed.
- You are *not allowed to use calculators*.

Good Luck!

Question	Pages	Parts	Points	Score
1. Recurrences	2 – 4	(a) – (d)	$4 \times 4 = 16$	
2. Asymptotic Bounds	5 – 7	(a) – (d)	$4 \times 4 = 16$	
3. Yet Another Sum	8 – 9	–	8	
4. Recursive Divide and Conquer	10 – 14	(a) – (b)	$7 + 8 = 15$	
5. Quicksort	15 – 18	(a) – (d)	$4 \times 5 = 20$	
Total	–	–	75	

NAME: _____

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Question 1. [16 Points] Recurrences. For each of the functions below either solve it using the Master Theorem (and state the case that applies) or state why the Master Theorem does not apply. Assume that all log's are of base 2 and $T(n) = \Theta(1)$ for $n \leq 1$.

(a) [4 Points] $T(n) = 4 \cdot T\left(\frac{n}{n^{\frac{1}{\log n}}}\right) + n$

(b) [4 Points] $T(n) = \left(\sum_{i=1}^{\log n} T\left(\frac{n}{2^i}\right)\right) + 2n$, where n is a power of 2

(c) [4 Points] $T(n) = 3 \cdot T\left(\frac{n}{2}\right) - n^2$

(d) [4 Points] $T(n) = \frac{1}{3} \cdot T(2n) - n^2$

Solution to Question 1.

(a) We know that $n^{\frac{1}{\log n}} = (2^{\log_2 n})^{\frac{1}{\log_2 n}} = 2^{\frac{\log_2 n}{\log_2 n}} = 2$.

Hence, the given recurrence reduces to the following:

$$T(n) = \begin{cases} \Theta(1) & \text{if } n \leq 1, \\ 4 \cdot T\left(\frac{n}{2}\right) + n & \text{otherwise.} \end{cases}$$

This recurrence is in the form potentially solvable using the Master Theorem, where $a = 4$, $b = 2$, and $f(n) = n$.

Now, the number of leaves in the recursion tree $= n^{\log_b a} = n^{\log_2 4} = n^2$.

Clearly, $f(n) = n = n^{2-\epsilon} = n^{\log_b a - \epsilon}$, where $\epsilon = 1 > 0$.

Hence, Case 1 of the Master Theorem applies, and we get: $T(n) = \Theta(n^{\log_b a}) = \Theta(n^2)$.

(b) We are given the following recurrence, where n is a power of 2.

$$T(n) = \left(\sum_{i=1}^{\log_2 n} T\left(\frac{n}{2^i}\right)\right) + 2n \tag{1}$$

Replacing n with $\frac{n}{2}$ in Equation 1, we get the following:

$$\begin{aligned}
T\left(\frac{n}{2}\right) &= \left(\sum_{i=1}^{\log_2 \frac{n}{2}} T\left(\frac{n}{2^i}\right) \right) + 2 \cdot \left(\frac{n}{2}\right) \\
\Rightarrow T\left(\frac{n}{2}\right) &= \left(\sum_{i=1}^{\log_2 n - 1} T\left(\frac{n}{2^{i+1}}\right) \right) + n \\
\Rightarrow T\left(\frac{n}{2}\right) &= \left(\sum_{i=2}^{\log_2 n} T\left(\frac{n}{2^i}\right) \right) + n
\end{aligned} \tag{2}$$

Now subtracting Equation 2 from Equation 1 we get:

$$\begin{aligned}
T(n) - T\left(\frac{n}{2}\right) &= T\left(\frac{n}{2}\right) + n \\
\Rightarrow T(n) &= 2 \cdot T\left(\frac{n}{2}\right) + n
\end{aligned}$$

Hence, the given recurrence is equivalent to the following:

$$T(n) = \begin{cases} \Theta(1) & \text{if } n \leq 1, \\ 2 \cdot T\left(\frac{n}{2}\right) + n & \text{otherwise.} \end{cases}$$

Comparing with the general form of the recurrences potentially solvable using the Master Theorem, we get: $a = 2$, $b = 2$, and $f(n) = n$.

The number of leaves in the recursion tree $= n^{\log_b a} = n^{\log_2 2} = n$.

Clearly, $f(n) = n = n^{\log_b a}$.

Hence, Case 2 of the Master Theorem applies, and we get: $T(n) = \Theta(n^{\log_b a} \log n) = \Theta(n \log n)$.

- (c) Comparing the given recurrence with the general form of the recurrences potentially solvable using the Master Theorem, we get: $f(n) = -n^2$ which is negative for $n > 0$. But the Master Theorem does not apply when $f(n)$ is negative. Hence, the given recurrence cannot be solved using the Master Theorem.

- (d) We are given the following.

$$\begin{aligned}
T(n) &= \frac{1}{3} \cdot T(2n) - n^2 \\
\Rightarrow 3 \cdot T(n) &= T(2n) - 3n^2 \\
\Rightarrow T(2n) &= 3 \cdot T(n) + 3n^2
\end{aligned} \tag{4}$$

Replacing n with $\frac{n}{2}$ in Equation 4, we get the following:

$$T(n) = 3 \cdot T\left(\frac{n}{2}\right) + \frac{3}{4} \cdot n^2$$

Hence, the given recurrence is equivalent to the following:

$$T(n) = \begin{cases} \Theta(1) & \text{if } n \leq 1, \\ 3 \cdot T\left(\frac{n}{2}\right) + \frac{3}{4} \cdot n^2 & \text{otherwise.} \end{cases}$$

Comparing with the general form of the recurrences potentially solvable using the Master Theorem, we get: $a = 3$, $b = 2$, and $f(n) = \frac{3}{4} \cdot n^2$.

The number of leaves in the recursion tree $= n^{\log_b a} = n^{\log_2 3}$.

Clearly, $f(n) = \Theta(n^2) = \Omega(n^{\log_2 3 + (2 - \log_2 3)}) = \Omega(n^{\log_b a + \epsilon})$, where $\epsilon = 2 - \log_2 3 = \log_2 4 - \log_2 3 = \log_2 \frac{4}{3} > 0$.

So, Case 3 of the Master Theorem may apply, but for it to apply the following regularity condition must also hold: $a \cdot f\left(\frac{n}{b}\right) \leq c \cdot f(n)$, where $c < 1$ is a constant.

Now, $a \cdot f\left(\frac{n}{b}\right) = 3 \cdot f\left(\frac{n}{2}\right) = 3 \cdot \frac{3}{4} \cdot \left(\frac{n}{2}\right)^2 = \frac{3}{4} \cdot \frac{3}{4} \cdot n^2 = \frac{3}{4} \cdot f(n) \leq c \cdot f(n)$, where $c = \frac{3}{4} < 1$.

So, the regularity condition also holds, and thus Case 3 of the Master Theorem applies.

Therefore, $T(n) = \Theta(f(n)) = \Theta(n^2)$.

Question 2. [16 Points] Asymptotic Bounds. For each of the following statements, state whether it is *True* or *False*. Justify each answer. Assume that all log's are of base 2.

(a) [4 Points] $\frac{1}{n} = \mathcal{O}\left(\frac{1}{n^2}\right)$

(b) [4 Points] $\log(n!) = \mathcal{O}(n \log n)$

(c) [4 Points] $n^{\log \log n} = \omega\left((\log n)^{\log n}\right)$

(d) [4 Points] $\sum_{i=0}^n \frac{1}{2^i} = \Theta\left(\sum_{i=0}^n \frac{1}{n^i}\right)$

Solution to Question 2.

(a) **False.**

We have $\lim_{n \rightarrow \infty} \frac{\frac{1}{n}}{\frac{1}{n^2}} = \lim_{n \rightarrow \infty} n = \infty$, which implies that $\frac{1}{n} = \omega\left(\frac{1}{n^2}\right)$ and so $\frac{1}{n} \neq \mathcal{O}\left(\frac{1}{n^2}\right)$.

(b) **True.**

We have $\log n! = \log\left(\prod_{k=1}^n k\right) \leq \log\left(\prod_{k=1}^n n\right) = \log(n^n) = n \log n$.

Hence, $\log n! = \mathcal{O}(n \log n)$.

(c) **False.**

We have

$$\begin{aligned} \bullet \quad n^{\log \log n} &= n^{\log_2 \log_2 n} = \left(2^{\log_2 n}\right)^{\log_2 \log_2 n} = 2^{(\log_2 n)(\log_2 \log_2 n)} \\ \bullet \quad (\log n)^{\log n} &= (\log_2 n)^{\log_2 n} = \left(2^{\log_2 \log_2 n}\right)^{\log_2 n} = 2^{(\log_2 n)(\log_2 \log_2 n)} \end{aligned}$$

Hence, $n^{\log \log n} = (\log n)^{\log n} \Rightarrow n^{\log \log n} = \Theta\left((\log n)^{\log n}\right)$, and so $n^{\log \log n} \neq \omega\left((\log n)^{\log n}\right)$.

(d) **True.**

We know that $1 + x + x^2 + x^3 + \dots + x^n = \frac{1-x^{n+1}}{1-x}$ holds for all nonnegative $x \neq 1$.

Hence, we have: $\lim_{n \rightarrow \infty} \frac{\sum_{i=0}^n \frac{1}{2^i}}{\sum_{i=0}^n \frac{1}{n^i}} = \lim_{n \rightarrow \infty} \frac{\frac{1 - \frac{1}{2^{n+1}}}{1 - \frac{1}{2}}}{1 + \frac{1}{n} + \frac{1}{n^2} + \frac{1}{n^3} + \dots + \frac{1}{n^n}} = \frac{\frac{1}{1 - \frac{1}{2}}}{1} = 2$, which implies that $\sum_{i=0}^n \frac{1}{2^i} = \Theta\left(\sum_{i=0}^n \frac{1}{n^i}\right)$.

Question 3. [8 Points] Yet Another Sum. Give the best upper bound you can derive on the worst-case running time of SUM-5 for an input of size n . You must show your analysis.

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SUM-5(  $n, A[1 : n]$  )
1.  $s \leftarrow 0$ 
2.  $d \leftarrow 1$ 
3.  $x \leftarrow \log_e \log_e n$ 
4. for  $i \leftarrow 1$  to  $n$  do
5.      $d \leftarrow d \times \left(\frac{i}{x}\right)$ 
6.      $j \leftarrow \lceil d \rceil$ 
7.     while  $j \leq n$  do
8.          $s \leftarrow s + A[j]$ 
9.          $j \leftarrow j + \lceil d \rceil$ 
10. return  $s$ 

```

You may find the following series useful (which holds for all real values of x):

$$e^x = \sum_{i=1}^{\infty} \frac{x^i}{i!} = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \dots \quad \dots \quad \dots,$$

where $e \approx 2.7182818$ is the base of natural logarithm.

Solution to Question 3.

- Lines 1–3 take $\Theta(1)$ time.
- The body of the loop in line 4 is executed n times (although line 4 itself is executed once more for a failed boundary check). Since each execution of lines 5–6 take $\Theta(1)$ time, lines 4–6 together take $\Theta(n)$ time overall.
- In the i^{th} iteration of the loop in line 4, $d = \frac{i!}{x^i}$. Hence, the body of the loop in line 7 is executed $1 + \left\lfloor \frac{n - \left\lceil \frac{i!}{x^i} \right\rceil}{\left\lceil \frac{i!}{x^i} \right\rceil} \right\rfloor$ times (again the boundary check requires an additional execution of line 7).

So, the body of the loop in line 7 is executed $\sum_{i=1}^n \left(1 + \left\lfloor \frac{n - \left\lceil \frac{i!}{x^i} \right\rceil}{\left\lceil \frac{i!}{x^i} \right\rceil} \right\rfloor \right) = n + \sum_{i=1}^n \left\lfloor \frac{n}{\left\lceil \frac{i!}{x^i} \right\rceil} - 1 \right\rfloor$ times in total, and line 7 itself is executed an additional n times (for failed boundary checks). Each execution of the loop body in lines 8–9 takes $\Theta(1)$ time. Now, since $\lfloor x \rfloor \leq x$ and $\lceil x \rceil \geq x$ for any $x \geq 0$, we have:

$$\begin{aligned}
n + \sum_{i=1}^n \left\lfloor \frac{n}{\left\lceil \frac{i!}{x^i} \right\rceil} - 1 \right\rfloor &\leq n + \sum_{i=1}^n \left(\frac{n}{\left\lceil \frac{i!}{x^i} \right\rceil} - 1 \right) \\
&= n + \sum_{i=1}^n \left(\frac{n}{\left\lceil \frac{i!}{x^i} \right\rceil} \right) - \sum_{i=1}^n 1 \\
&= n + \sum_{i=1}^n \left(\frac{n}{\left\lceil \frac{i!}{x^i} \right\rceil} \right) - n \\
&= \sum_{i=1}^n \left(\frac{n}{\left\lceil \frac{i!}{x^i} \right\rceil} \right) \\
&\leq \sum_{i=1}^n \left(\frac{n}{\frac{i!}{x^i}} \right) \\
&\leq n \cdot \sum_{i=1}^n \left(\frac{x^i}{i!} \right) \\
&< n \cdot \sum_{i=1}^{\infty} \left(\frac{x^i}{i!} \right) \\
&< n \cdot e^x \\
&= n \cdot e^{\log_e \log_e n} \\
&= n \cdot \log_e n \\
&= \mathcal{O}(n \log n)
\end{aligned}$$

Hence, lines 7–9 take $\mathcal{O}(n \log n)$ time overall.

- Line 10 takes $\Theta(1)$ time.

Therefore, SUM-5 takes $\Theta(1) + \Theta(n) + \mathcal{O}(n \log n) + \Theta(1) = \mathcal{O}(n \log n)$ time overall.

Question 4. [15 Points] Recursive Divide and Conquer. The `REC-INPLACE-FFT` function given below is an in-place version of the Fast Fourier Transform (FFT) algorithm. It is based on recursive divide and conquer and uses the `UNSHUFFLE` function (also given below) as a subroutine. The `UNSHUFFLE` function, too, uses recursive divide and conquer.

$$\text{REC-INPLACE-FFT}(A, n)$$

Input: An array $A[1 : n]$ of n complex numbers, where $n = 2^k$ for some integer $k \geq 0$.

Output: The Discrete Fourier Transform (DFT) of the input array computed in place in $A[1 : n]$.

Algorithm:

1. **if** $n > 1$ **then**
2. $\text{UNSHUFFLE}(A, n)$
3. Let A_L denote $A[1 : \frac{n}{2}]$ (i.e., the left half of A) and
 let A_R denote $A[\frac{n}{2} + 1 : n]$ (i.e., the right half of A)
4. $\text{REC-INPLACE-FFT}(A_L, \frac{n}{2})$
5. $\text{REC-INPLACE-FFT}(A_R, \frac{n}{2})$
6. $y \leftarrow \cos\left(\frac{2\pi}{n}\right) + i \cdot \sin\left(\frac{2\pi}{n}\right)$ $\{i = \sqrt{-1}\}$
7. $z \leftarrow 1$
8. **for** $j \leftarrow 1$ **to** $\frac{n}{2}$ **do**
9. $l \leftarrow A_L[j]$
10. $r \leftarrow A_R[j]$
11. $A_L[j] \leftarrow l + z \cdot r$
12. $A_R[j] \leftarrow l - z \cdot r$
13. $z \leftarrow z \cdot u$

$$\text{UNSHUFFLE}(A, n)$$

Input: An array $A[1 : n]$ of n complex numbers, where $n = 2^k$ for some integer $k \geq 0$.

Output: The entries of $A[1 : n]$ rearranged in place so that all entries located at odd (resp. even) indices of the input array are moved to the left (resp. right) half of the array without changing the order in which they appeared in the input array.

Algorithm:

1. **if** $n > 2$ **then**
2. Let A_L denote $A[1 : \frac{n}{2}]$ (i.e., the left half of A) and
 let A_R denote $A[\frac{n}{2} + 1 : n]$ (i.e., the right half of A)
3. UNSHUFFLE($A_L, \frac{n}{2}$)
4. UNSHUFFLE($A_R, \frac{n}{2}$)
5. **for** $j \leftarrow 1$ **to** $\frac{n}{4}$ **do**
6. $A_L[\frac{n}{4} + j] \leftrightarrow A_R[j]$ $\{swap\}$

- (a) [7 Points] Let $T'(n)$ denote the worst-case running time of UNSHUFFLE on an array of size n . What is the cost of the **divide** step of the algorithm? What is the cost of the **combine** step? Write a recurrence relation for $T'(n)$ and solve it.
- (b) [8 Points] Let $T(n)$ be the worst-case running time of REC-INPLACE-FFT on an array of size n . What is the cost of the **divide** step of the algorithm? What is the cost of the **combine** step? Write a recurrence relation for $T(n)$ and solve it.

Solution to Question 4.

- (a) The UNSHUFFLE algorithm does not have an explicit divide step, and so the cost (or time complexity) of the divide step is $f'_d(n) = 0$.

The loop in lines 5–6 represent the combine step. Line 5 is executed $\frac{n}{4} + 1$ times taking $\Theta(1)$ time per execution, and the loop body in line 6 is executed $\frac{n}{4}$ times with $\Theta(1)$ time per execution. Hence, the total cost of executing the loop in lines 5–6 is $\Theta(n)$, and therefore the cost of the combine step is $f'_c(n) = \Theta(n)$.

Thus, the total cost of divide and combine is $f'(n) = f'_d(n) + f'_c(n) + \Theta(1) = \Theta(n)$, where we have added $\Theta(1)$ to account for the $\Theta(1)$ time taken by instructions not directly related to the divide and combine steps (e.g., the **if** condition in line 1).

The base case is reached when $n \leq 2$ and it takes $\Theta(1)$ time (e.g., to execute the **if** condition in line 1, return from the recursive call, etc.).

When $n > 2$ the algorithm makes two recursive calls (in lines 3 and 4) to solve two subproblems of half the size.

Therefore, the recurrence for $T'(n)$ can be written as follows.

$$T'(n) = \begin{cases} \Theta(1) & \text{if } n \leq 2, \\ 2 \cdot T'(\frac{n}{2}) + \Theta(n) & \text{otherwise.} \end{cases}$$

The number of leaves in the recursion tree of $T'(n)$ is $\Theta(n^{\log_2 2}) = \Theta(n) = f'(n)$. So, Case 2 of the Master Theorem applies and hence, $T'(n) = \Theta(f'(n) \log n) = \Theta(n \log n)$.

- (b) The divide step consists of a call to UNSHUFFLE(A , n) in line 2, and so the cost of the divide step is, $f_d(n) = \Theta(n \log n)$ (from part (a)).

The combine step is in lines 6–13. Lines 6–7 take $\Theta(1)$ time. Line 8 is executed $\frac{n}{2} + 1$ times with the loop body in lines 9–13 is executed $\frac{n}{2}$ times taking $\Theta(1)$ time per execution. So, the cost of the combine step is, $f_c(n) = \Theta(n)$.

Hence, the total cost of divide and combine (including $\Theta(1)$ cost for instructions not directly related to those two steps) is, $f(n) = f_d(n) + f_c(n) + \Theta(1) = \Theta(n \log n) + \Theta(n) + \Theta(1) = \Theta(n \log n)$.

The base case is reached when $n \leq 1$ and takes $\Theta(1)$ time. The algorithm makes two recursive calls (in lines 4–5) to solve two subproblems of half the size.

Therefore, the recurrence for $T(n)$ can be written as follows.

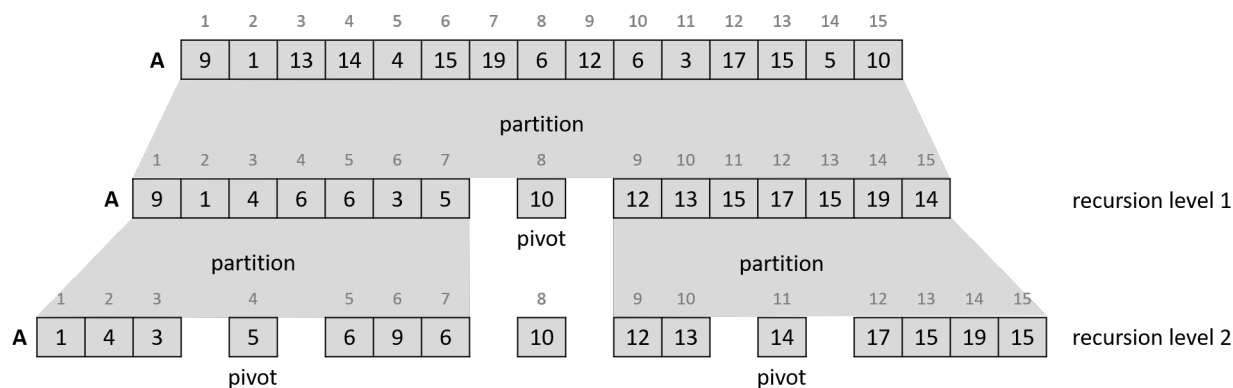
$$T(n) = \begin{cases} \Theta(1) & \text{if } n \leq 1, \\ 2 \cdot T\left(\frac{n}{2}\right) + \Theta(n \log n) & \text{otherwise.} \end{cases}$$

The number of leaves in the recursion tree of $T(n)$ is $\Theta(n^{\log_2 2}) = \Theta(n)$. But $f(n) = \Theta(n \log n)$ which is $\Theta(\log n)$ factor larger than the number of leaves. So, Case 2 (the extended version of it) of the Master Theorem applies, and hence $T(n) = \Theta(f(n) \log n) = \Theta(n \log^2 n)$.

Question 5. [20 Points] Quicksort.

Recall the in-place partitioning algorithm we learned in the class which is used by quicksort to partition the numbers in the array or subarray it is working on around the pivot. We always chose the last element of the given array/subarray as the pivot.

Given an array $A[1 : 15]$ as input, the following example shows the state of A after partitioning in the first two levels of recursion of quicksort.



You can describe the state of A after partitioning in recursion level 1 as:

$[9, 1, 4, 6, 6, 3, 5] \ 10 \ [12, 13, 15, 17, 15, 19, 14]$

Then after partitioning in recursion level 2 as:

$[[1, 4, 3] \ 5 \ [6, 9, 6]] \ 10 \ [[12, 13] \ 14 \ [17, 15, 19, 15]]$

Now, given the following array $A[1 : 16]$ of numbers, you will have to describe its state after partitioning at recursion levels 1, 2, 3 and 4.

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
A	97	22	31	38	4	71	55	12	64	88	33	17	48	92	8	43

- [5 Points] Describe the state of A after partitioning at recursion level 1.
- [5 Points] Describe the state of A after partitioning at recursion level 2.
- [5 Points] Describe the state of A after partitioning at recursion level 3.
- [5 Points] Describe the state of A after partitioning at recursion level 4.

Solution to Question 5.

$$(a) \ [22, 31, 38, 4, 12, 33, 17, 8] \ 43 \ [88, 71, 55, 48, 92, 97, 64]$$

$$(b) \ [[4] \ 8 \ [38, 22, 12, 33, 17, 31]] \ 43 \ [[55, 48] \ 64 \ [71, 92, 97, 88]]$$

$$(c) \ [[4] \ 8 \ [[22, 12, 17] \ 31 \ [38, 33]]] \ 43 \ [[48 \ [55]] \ 64 \ [[71] \ 88 \ [97, 92]]]$$

$$(d) \ [[4] \ 8 \ [[[12] \ 17 \ [22]] \ 31 \ [33 \ [38]]]] \ 43 \ [[48 \ [55]] \ 64 \ [[71] \ 88 \ [[92] \ 97]]]$$